We will begin the class at 2:05 p.m.

Lecture 2: Inverse Trigonometric Functions

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

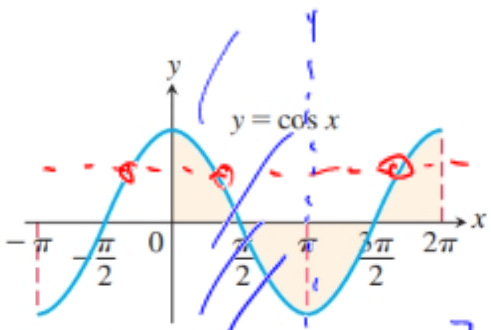
Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

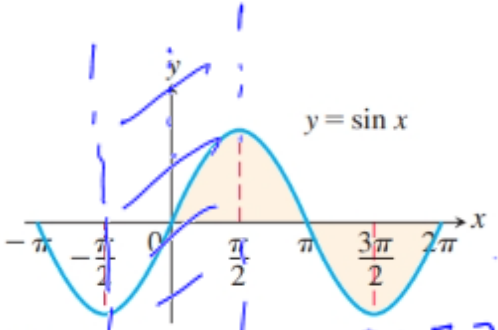
- ☐ Find and calculate the inverse trigonometric function.
- ☐ Find the derivative of the inverse trigonometric function.
- ☐ Perform basic integration using integration formula involving inverse trigonometric function.

Trigonometric Functions



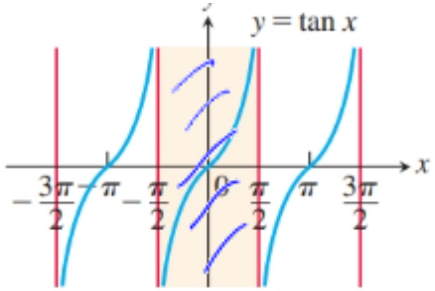
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(a)



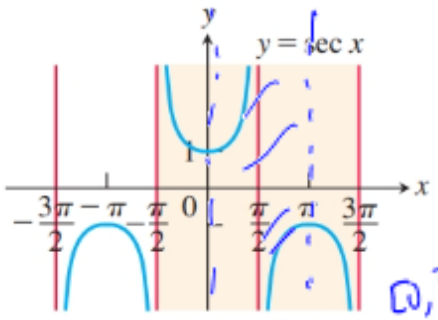
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(b)



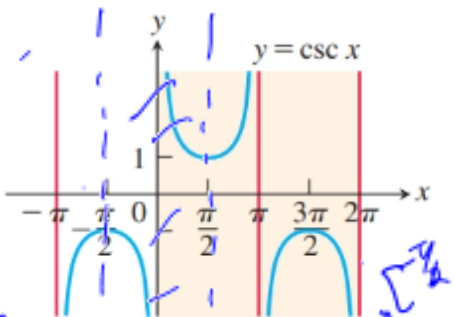
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $-\infty < y < \infty$
Period: π

(c)



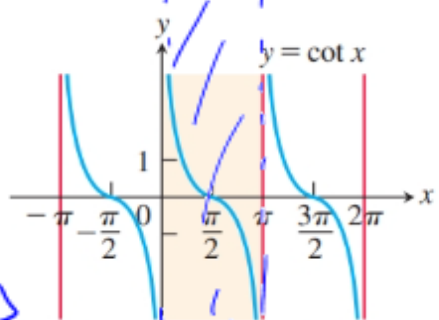
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π

(d)



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π

(e)

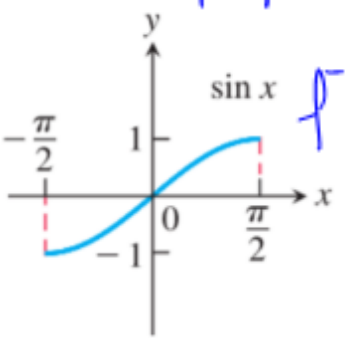


Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $-\infty < y < \infty$
Period: π

(f)

Generally, the trigonometric functions are not 1-1 function. However, as we learned in previous lecture, we can restrict their domain to make them 1-1 and hence have an inverse.

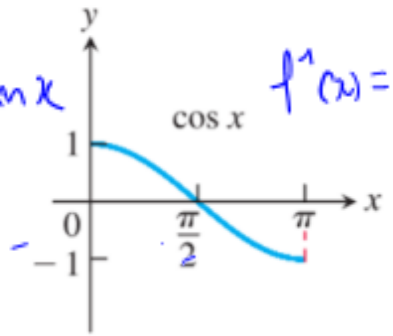
Trigonometric Functions with restricted domain



1−1

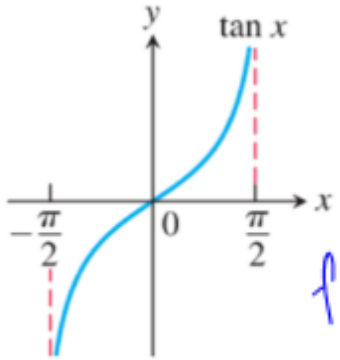
$f^{-1}(x) = \arcsin x$

$y = \sin x$
Domain: $[-\pi/2, \pi/2]$
Range: $[-1, 1]$



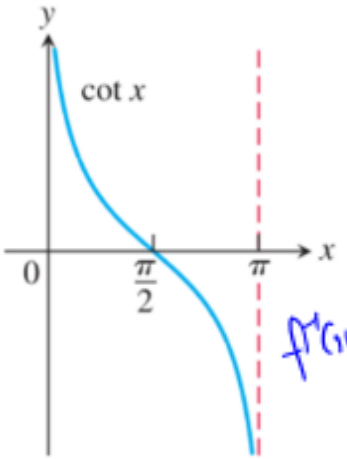
$f^{-1}(x) = \arccos x$

$y = \cos x$
Domain: $[0, \pi]$
Range: $[-1, 1]$



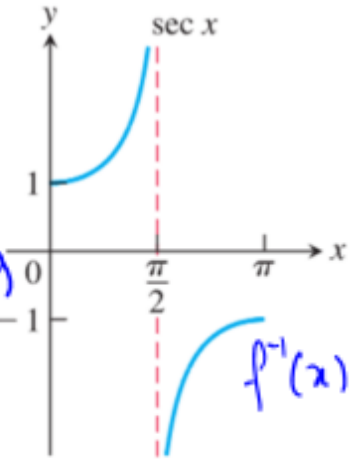
$f^{-1}(x) = \arctan x$

$y = \tan x$
Domain: $(-\pi/2, \pi/2)$
Range: $(-\infty, \infty)$



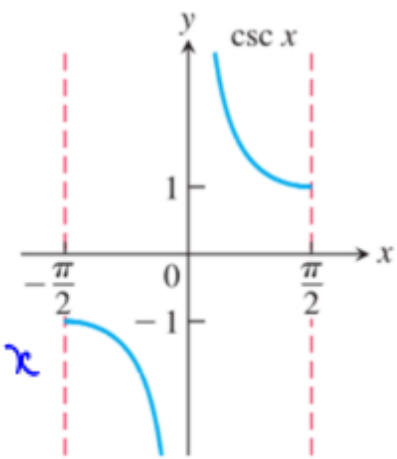
$f^{-1}(x) = \operatorname{arccot} x$

$y = \cot x$
Domain: $(0, \pi)$
Range: $(-\infty, \infty)$



$f^{-1}(x) = \operatorname{arcsec} x$

$y = \sec x$
Domain: $[0, \pi/2) \cup (\pi/2, \pi]$
Range: $(-\infty, -1] \cup [1, \infty)$

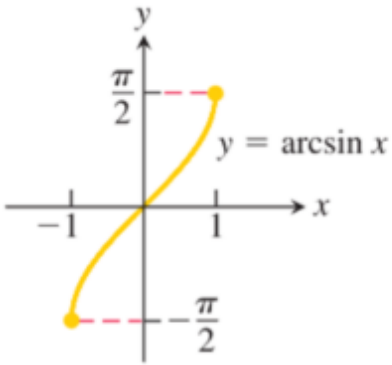


$f^{-1}(x) = \operatorname{arccsc} x$

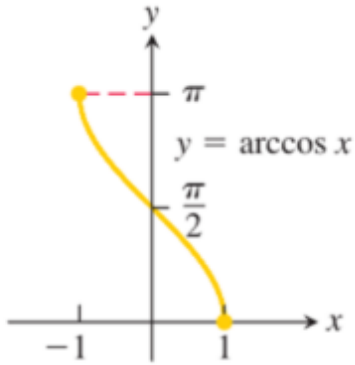
$y = \csc x$
Domain: $[-\pi/2, 0) \cup (0, \pi/2]$
Range: $(-\infty, -1] \cup [1, \infty)$

Inverse Trigonometric Functions

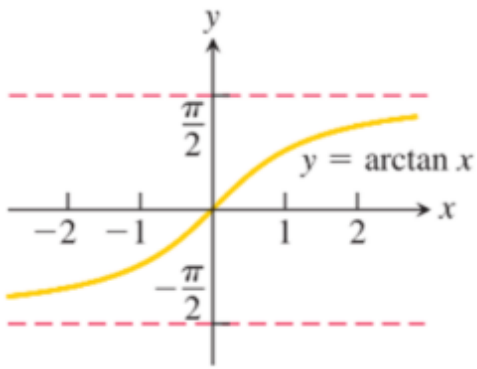
Domain: $-1 \le x \le 1$
Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$



Domain: $-1 \le x \le 1$
Range: $0 \le y \le \pi$



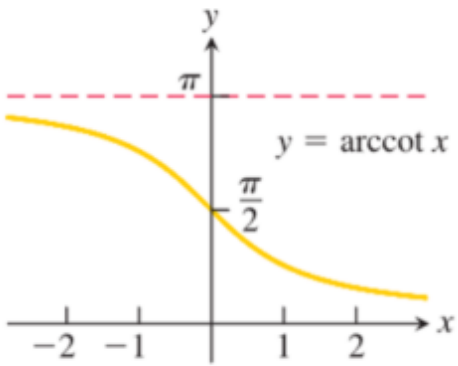
Domain: $-\infty < x < \infty$
Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$



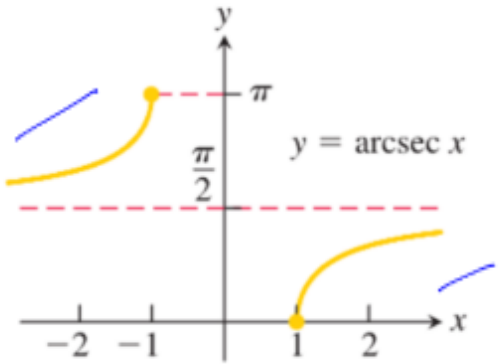
reflexion on $y=x$.

$\Leftarrow$ odd function

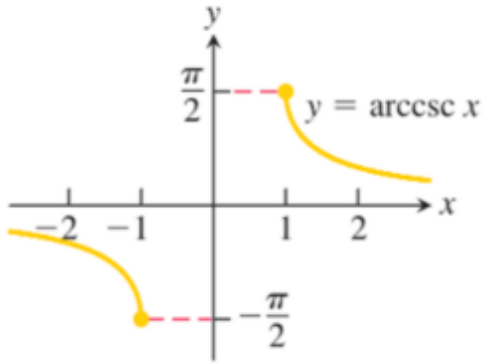
Domain: $-\infty < x < \infty$
Range: $0 < y < \pi$



Domain: $x \le -1$ or $x \ge 1$
Range: $0 \le y \le \pi, y \ne \frac{\pi}{2}$



Domain: $x \le -1$ or $x \ge 1$
Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}, y \ne 0$



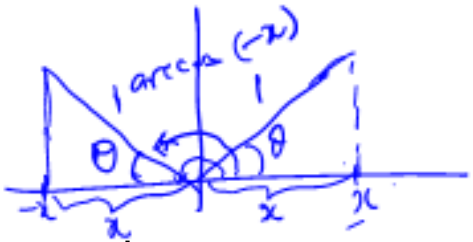
Inverse Trigonometric Functions

These properties of inverse trigonometric functions can be useful

- $\arcsin(-x) = -\arcsin x$

  symmetry about the origin $\Rightarrow$ odd function $\left[f(-x) = -f(x) \right]$

- $\arccos(-x) = \pi - \arccos x$



$\cos\theta = x$

$\arccos(-x) + \arccos x = \pi$

$\arccos(-x) = \pi - \arccos x$

- $\arctan(-x) = -\arctan x$

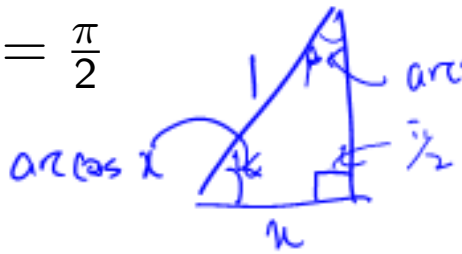
  odd function $\left[f(-x) = -f(x) \right]$

$\alpha + \beta = \pi/2$

- $\arcsin(x) + \arccos(x) = \frac{\pi}{2}$

$\cos\alpha = \frac{x}{1}$

$\sin\beta = \frac{x}{1}$



$\arccos x + \arcsin x = \pi/2$

Inverse Trigonometric Functions

Example: Evaluate

- $\arcsin\left(\frac{\sqrt{3}}{2}\right)$

  $= \pi/3$

  Let $\arcsin\frac{\sqrt{3}}{2} = \theta$, $\sin\theta = \frac{\sqrt{3}}{2}$

  As we know, $\sin\pi/3 = \frac{\sqrt{3}}{2}$ and $\pi/3 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ ← range of arcsin

- $\arccos\left(-\frac{1}{2}\right)$

  $= \frac{2\pi}{3}$

  Let $\arccos\frac{1}{2} = \theta$, $\cos\theta = -\frac{1}{2}$

  As we know, $\cos\frac{2\pi}{3} = -\frac{1}{2}$ and $\frac{2\pi}{3} \in [0, \pi]$

- $\arcsin\left(\sin\left(\frac{3\pi}{4}\right)\right) = \frac{3\pi}{4}$? ✗

  Range $\arcsin(x) = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

  θ

  "$\arcsin\left(\sin\frac{\pi}{4}\right) = \frac{\pi}{4}$ and $\frac{3\pi}{4} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

- $\cos\left(\arcsin\left(\frac{2}{7}\right)\right)$

  $= \frac{\sqrt{45}}{7} = \frac{3\sqrt{5}}{7}$

  Let $\arcsin\frac{2}{7} = \theta$, $\sin\theta = \frac{2}{7}$

  $\sqrt{7^2-2^2}$

  $\cos\theta = \frac{\sqrt{7^2-2^2}}{7}$

  $= \frac{\sqrt{45}}{7}$

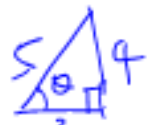
- $\sin(2\arccos\left(\frac{3}{5}\right))$

  $= \frac{24}{25}$

  Let $\theta = \arccos\left(\frac{3}{5}\right)$

  $\sin 2\theta = 2\sin\theta\cos\theta$

  $= 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right) = \frac{24}{25}$

Inverse Trigonometric Functions

Note that

$$\sin(\arcsin(x)) = \cos(\arccos(x)) = \tan(\arctan(x)) = x$$

is true for all x,

but

$$\arcsin(\sin(x)) = \arccos(\cos(x)) = \arctan(\tan(x)) = x$$

is only true for some x.

only true when x is in the range of the arc function.

Derivative of Inverse Trigonometric Function

1. $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}} \quad (|x| < 1)$

2. $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$

3. $\frac{d}{dx}(\operatorname{arcsec} x) = \frac{1}{|x|\sqrt{x^2-1}} \quad (|x| > 1)$

4. $\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}} \quad (|x| < 1)$

5. $\frac{d}{dx}(\operatorname{arccot} x) = -\frac{1}{1+x^2}$

6. $\frac{d}{dx}(\operatorname{arccsc} x) = -\frac{1}{|x|\sqrt{x^2-1}} \quad (|x| > 1)$

Derivative of Inverse Trigonometric Function

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad |x|<1$$

$\sin^2 x + \cos^2 x = 1$

(M1) $\left[\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}\right]$

Let $y = \arcsin x$ $\quad -\pi/2 \le y \le \pi/2$ ← range of arcsin

S | A

T | C

$x = \sin y$

$\frac{dx}{dy} = \cos y = \pm\sqrt{1-\sin^2 y} = \pm\sqrt{1-x^2}$

$= \sqrt{1-x^2}, \quad -\frac{\pi}{2} \le y \le \pi/2$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$ $\quad (1-x^2 > 0 \Rightarrow |x|<1)$

(M2) $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

Let $f(x) = \sin x$ $\quad f^{-1}(x) = \arcsin x$

$f'(x) = \cos x$

$\frac{d}{dx}(\arcsin x) = \frac{1}{f'(\arcsin x)} = \frac{1}{\cos(\arcsin x)}$

Derivative of Inverse Trigonometric Function

$$\frac{d}{dx}(\operatorname{arcsec} x) = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x|>1$$

$$= \frac{1}{\sqrt{1-\sin^2(\operatorname{arc\,sec} x)}} \qquad \sin(\operatorname{arc\,sec} x) = x$$

$$= \frac{1}{\sqrt{1-x^2}}$$

(M1) Let $y = \operatorname{arc\,sec} x$ $\quad 0 \le y \le \pi, \ y \neq \pi/2$

$$x = \sec y$$

$$\frac{dx}{dy} = \sec y \tan y$$

$$= \sec y \left(\pm\sqrt{\sec^2 y - 1}\right)$$

$$= \pm x\sqrt{x^2-1}$$

$$\frac{dy}{dx} = \frac{1}{\pm x\sqrt{x^2-1}} \qquad \begin{matrix} \pm x: \ +ve \\ >0 \end{matrix}, \quad \sqrt{x^2-1} \ge 0, \quad |x|>1$$

$$= \frac{1}{|x|\sqrt{x^2-1}}, \quad |x|>1$$

Exercise: Try to show the other derivatives.

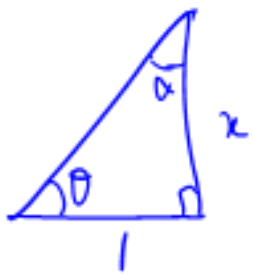
Derivative of Inverse Trigonometric Function

Inverse Function-Inverse Cofunction Identities

$$\arccos x = \frac{\pi}{2} - \arcsin x$$

$$\operatorname{arccot} x = \frac{\pi}{2} - \arctan x$$

$$\operatorname{arccsc} x = \frac{\pi}{2} - \operatorname{arcsec} x \quad \leftarrow \text{exercise.}$$



$\tan\theta = \frac{x}{1} \Rightarrow \theta = \arctan x$

$\tan\alpha = \frac{1}{x}$

$\cot\alpha = x \Rightarrow \alpha = \operatorname{arccot} x$

$\alpha + \theta = \pi/2 \Rightarrow \arctan x + \operatorname{arccot} x = \pi/2$.

Derivative of Inverse Trigonometric Function

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}, \quad |x|<1$$

$$\arccos x = \pi/2 - \arcsin x$$

$$\frac{d}{dx}(\arccos x) = \frac{d}{dx}\left(\underbrace{\pi/2}_{\text{constant}} - \arcsin x\right)$$

$$= 0 - \frac{1}{\sqrt{1-x^2}}$$

$$= -\frac{1}{\sqrt{1-x^2}}, \quad |x|<1$$

Derivative of Inverse Trigonometric Function

Example: Evaluate

1. $\frac{d}{dx}(\arcsin x^2)$ $= \frac{1}{\sqrt{1-(x^2)^2}} \frac{d}{dx}(x^2) = \frac{2x}{\sqrt{1-x^4}}$ (chain rule.)
2. $\frac{d}{dx}(\frac{1}{\arctan x})$ $= \frac{d}{dx}[\arctan(x)]^{-1} = -\frac{1}{\arctan^2(x)}\left[\frac{d}{dx}(\arctan x)\right]$
3. $\frac{d}{dx}(\operatorname{arcsec}(5x^4))$ $= \frac{-1}{\arctan^2(x)}\left[\frac{1}{1+x^2}\right]$
4. $\frac{d}{dx}(\arcsin(3x^4+5x+2))$ $= \frac{-1}{(\arctan^2(x))(1+x^2)}$
5. $\frac{d}{dx}(\ln(x)\arccos(x))$
6. $\frac{d}{dx}(\frac{2x}{\arcsin x})$

3. $\frac{d}{dx}\left(\operatorname{arcsec}\left(5x^4\right)\right) = \frac{1}{\left|5x^4\right|\sqrt{(5x^4)^2-1}}\frac{d}{dx}\left(5x^4\right)$

$5x^4 > 0$

$= \frac{20x^3}{5x^4\sqrt{25x^8-1}}$

$= \frac{4}{x\sqrt{25x^8-1}}$

4. $\frac{d}{dx}\left[\arcsin\left(3x^4+5x+2\right)\right]$

$= \frac{1}{\sqrt{1-(3x^4+5x+2)^2}}\frac{d}{dx}\left(3x^4+5x+2\right)$

$= \frac{12x^3+5}{\sqrt{1-(3x^4+5x+2)^2}}$

5. $\frac{d}{dx}\left[\ln x \arccos x\right] = \left(\frac{d}{dx}\ln x\right)(\arccos x) + \left(\frac{d}{dx}\arccos x\right)(\ln x)$ product rule.

$$= \frac{1}{x}\arccos x + \left(\frac{-1}{\sqrt{1-x^2}}\right)\ln x$$

$$= \frac{1}{x}\arccos x - \frac{\ln x}{\sqrt{1-x^2}}$$

6. $\frac{d}{dx}\left(\frac{2x}{\arcsin x}\right) = \frac{\left[\frac{d}{dx}(2x)\right](\arcsin x) - \left[\frac{d}{dx}(\arcsin x)\right](2x)}{(\arcsin x)^2}$ quotient rule.

$$= \frac{2\arcsin x - \frac{1}{\sqrt{1-x^2}}(2x)}{(\arcsin x)^2}$$

$$= \frac{2}{\arcsin x} - \frac{2x}{\left(\sqrt{1-x^2}\right)(\arcsin x)^2}$$

Integration Formulas

Integrals evaluated with inverse trigonometric functions

The following formulas hold for any constant $a > 0$.

1. $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$ (Valid for $x^2 < a^2$)

Apply chain rule for $\frac{d}{dx}\left(\arcsin\frac{x}{a}\right)$

$a=1$, $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$ $\left[\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}\right]$

2. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$ (Valid for all x)

3. $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\left|\frac{x}{a}\right| + C$ (Valid for $|x| > a > 0$)

Integration Formulas

Example: Evaluate

1. $\displaystyle\int_{\sqrt{2}/2}^{\sqrt{3}/2} \frac{dx}{\sqrt{1-x^2}}$ $= \left[\arcsin x\right]_{\sqrt{2}/2}^{\sqrt{3}/2} = \arcsin\left(\frac{\sqrt{3}}{2}\right) - \arcsin\frac{\sqrt{2}}{2}$
$= \pi/3 - \pi/4 = \pi/12$

2. $\displaystyle\int \frac{dx}{\sqrt{3-4x^2}}$ (2) Let $a^2 = 3 \Rightarrow a = \sqrt{3}$
Let $u^2 = 4x^2 \Rightarrow u = 2x$
$du = 2dx$

3. $\displaystyle\int \frac{dx}{\sqrt{e^{2x}-6}}$ $\displaystyle\int \frac{dx}{\sqrt{3-4x^2}} = \int \frac{\frac{1}{2}du}{\sqrt{a^2-u^2}}$

4. $\displaystyle\int \frac{dx}{\sqrt{4x-x^2}}$ $= \frac{1}{2}\arcsin\left(\frac{u}{a}\right) + C$

5. $\displaystyle\int \frac{dx}{4x^2+4x+2}$ $= \frac{1}{2}\arcsin\left(\frac{2x}{\sqrt{3}}\right)$

(3) $\int \frac{dx}{e^{2x}-6}$

Let $a^2 = 6$ and $u^2 = e^{2x}$

$\Rightarrow a = \sqrt{6}$ $\Rightarrow u = e^x$

$du = e^x dx$

$dx = \frac{1}{u} du$

$= \int \frac{du}{u\sqrt{u^2-a^2}}$

$= \frac{1}{a} \operatorname{arcsec}\left|\frac{u}{a}\right| + C$

$= \frac{1}{\sqrt{6}} \operatorname{arcsec}\left|\frac{e^x}{\sqrt{6}}\right| + C$

④ $\int \frac{dx}{\sqrt{4x - x^2}} = \int \frac{dx}{\sqrt{-(x^2 - 4x)}}$

Do completing the square

$= \int \frac{dx}{\sqrt{-[(x-2)^2 - 4]}}$ ← completing the square

$= \int \frac{dx}{\sqrt{4 - (x-2)^2}}$

Let $a^2 = 4$ and $u^2 = (x-2)^2$

$a = 2$ and $u = x - 2$

$du = dx$

$= \int \frac{du}{\sqrt{a^2 - u^2}}$

$= \arcsin\left(\frac{u}{a}\right) + C$

$= \arcsin\left(\frac{x-2}{2}\right) + C$

⑤ $\int \frac{dx}{4x^2+4x+2} = \int \frac{dx}{4(x^2+x)+2}$

$= \int \frac{dx}{4\left[(x+\frac{1}{2})^2-\frac{1}{4}\right]+2}$

$= \int \frac{dx}{4(x+\frac{1}{2})^2+1}$

2^2

$= \int \frac{dx}{(2x+1)^2+1}$

Let $a^2=1$, $u^2=(2x+1)^2$

$a=1$ $u=2x+1$

$du=2dx$

$= \int \frac{\frac{1}{2}du}{a^2+u^2}$

$= \frac{1}{2}\left(\frac{1}{a}\right)\arctan\left(\frac{u}{a}\right)+C$

$= \frac{1}{2}\arctan(2x+1)+C$

Integration Formulas

Additional example: Evaluate

1. $\int \frac{dx}{\sqrt{4-9x^2}}$ ← exercise

2. $\int_0^1 \frac{dx}{x^2+4x+13}$ ← exercise

3. $\int \frac{4-x}{\sqrt{16-x^2}}dx = \int \frac{4}{\sqrt{16-x^2}}dx - \int \frac{x}{\sqrt{16-x^2}}dx$

$$\int \frac{4}{\sqrt{16-x^2}}\,dx = 4\int \frac{1}{\sqrt{16-x^2}}\,dx$$

4^2

$$= 4 \arcsin\left(\frac{x}{4}\right)$$

$$\int \frac{x}{\sqrt{16-x^2}}\,dx = \int \frac{-\frac{1}{2}\,du}{\sqrt{u}}$$

Let $u = 16-x^2$

$du = -2x\,dx$

$x\,dx = \frac{-1}{2}\,du$

$$= -\frac{1}{2}(2\sqrt{u})$$

$$= -\sqrt{u} = \sqrt{16-x^2}$$

$$\int \frac{4-x}{\sqrt{16-x^2}}\,dx = 4\arcsin\left(\frac{x}{4}\right) + \sqrt{16-x^2} + C.$$